

Data Cleaning & Essential Functions Assignment

Q1. What is data cleaning, and why is it important in data analysis?

What are the potential consequences of analysing unclean or messy data?

Explain the common steps involved in cleaning and organizing data.

Ans- Data cleaning is the process of identifying and correcting errors, inconsistencies, missing values, and duplicate records in a dataset to improve its quality. It prepares raw data for accurate and meaningful analysis.

Data cleaning is important because data analysis is only as reliable as the data used. Clean data ensures accurate results, reliable insights, and better decision-making. Without proper cleaning, analysis can produce misleading outcomes.

Consequences of Analysing Unclean or Messy Data

Analysing unclean data can have serious negative effects, such as:-

- 1.) **Incorrect Results** Errors, duplicates, or missing values can distort calculations and trends.
- 2.) **Poor Decision-Making** Business decisions based on inaccurate data may lead to wrong strategies and losses.
- 3.) **Misleading Insights** Patterns and relationships identified may be false or incomplete.
- 4.) **Reduced Credibility** Reports and findings may be unreliable, reducing trust in the analysis.
- 5.) **Increased Time and Cost** More time is spent fixing errors later instead of focusing on analysis.

Common Steps Involved in Cleaning and Organizing Data

- 1.) **Removing Duplicate Data** Identifying and deleting repeated records to avoid over-representation.
- 2.) **Handling Missing Values** Filling missing data using appropriate methods (mean, median, default values) or removing incomplete records.
- 3.) **Correcting Errors and Inconsistencies** Fixing spelling mistakes, incorrect formats, and inconsistent entries (e.g., “NY” vs “New York”).
- 4.) **Standardizing Data Formats** Ensuring uniform formats for dates, numbers, and text.
- 5.) **Validating Data** Checking that values fall within acceptable ranges and follow defined rules.

6.) Organizing and Structuring Data Sorting, categorizing, and arranging data into a logical structure suitable for analysis.

Q2. How would you sort the following dataset first by "Department" (A-Z) and then by "Salary" (Largest to Smallest)? Write a step-by-step approach.

Employee	Department	Salary
Sonu	IT	4000
Pranav	HR	5000
Rahul	IT	2500

Ans- To sort the given dataset first by Department (A–Z) and then by Salary (Largest to Smallest) in Excel, follow these steps:

- 1.) Select the entire dataset, including column headings.
- 2.) Go to the Data tab in Excel.
- 3.) Click on Sort to open the Sort dialog box.
- 4.) In the Sort by field, select Department and choose A to Z.
- 5.) Click Add Level.
- 6.) In the “Then by field”, select Salary and choose Largest to Smallest.
- 7.) Click OK to apply the sorting.

After this the data will be the data will be arranged alphabetically by Department, and within each department, employees will be ordered from highest to lowest salary.

Q3. Explain the use of text functions such as TRIM, LEFT, RIGHT, MID, and CONCAT in data cleaning.

Ans- Text functions in Excel are commonly used in data cleaning to correct and organize text data Some of them are as follows:

- **TRIM**
TRIM removes extra spaces from text, leaving only single spaces between words. It helps clean data copied from other sources where unwanted spaces may exist.
- **LEFT**
LEFT extracts a specified number of characters from the beginning of a text string. It is useful when you need part of a text, such as extracting a code or initials.
- **RIGHT**
RIGHT extracts characters from the end of a text string. It is often used to get suffixes, years, or last digits from a value.
- **MID**
MID extracts text from the middle of a string based on a starting position and length. It helps separate useful information from combined text.
- **CONCAT**
CONCAT combines text from multiple cells into one single text string. It is useful for merging first name and last name or rebuilding cleaned data.

Q4. What is the role of date functions like TODAY in managing datasets?

Ans- Date functions such as TODAY play an important role in managing and analyzing datasets by working with current dates automatically.

The TODAY() function returns the current date and updates automatically whenever the worksheet is opened or recalculated. It is commonly used to track deadlines, calculate durations, and monitor time-based data.

In data management, TODAY helps in:

Calculating age, tenure, or number of days between dates

Identifying overdue tasks or expired records

Creating dynamic reports that always reflect the current date

Comparing past and future dates easily

Overall, date functions like TODAY make datasets more dynamic, accurate, and useful for time-based analysis.

Q5. Apply Data Validation to restrict Quantity values to only whole numbers between 1 and 10. Configure an input message that appears when a user selects a cell in the "Quantity" column explaining: "Please enter a whole number between 1 and 10."

Set up an error alert message that triggers if the user enters a number less than 1 or greater than 10, showing:

"Invalid input! The quantity must be a whole number between 1 and 10."

Write a step-by-step approach for this question

Customer Name	Product Name	Category	Quantity	Unit Price (\$)
Jane Smith	Shoes	Electronics		81
Isabella Moore	Laptop	Electronics		121
Daniel Davis	Sofa	Clothing		239
Alex Moore	Shoes	Electronics		500
Michael Johnson	Table Lamp	Home Decor		423
Daniel Johnson	Backpack	Electronics		160
Isabella Davis	Headphones	Electronics		348
Jane Davis	Headphones	Electronics		152
Alex Wilson	T-shirt	Home Decor		369

Ans- To restrict the Quantity values to whole numbers between 1 and 10 and display the required messages in Excel, follow these steps:

Step 1: Select the Quantity column

- Select all the cells under the Quantity column (for example, D2:D10).

Step 2: Open Data Validation

1. Go to the Data tab.
2. Click on Data Validation in the Data Tools group.

Step 3: Set validation rules

1. In the Settings tab:
 - Allow → select Whole number
 - Data → select between
 - Minimum → enter 1
 - Maximum → enter 10

Step 4: Add Input Message

1. Click the Input Message tab.
2. Enter:
 - Title: Quantity Input
 - Input Message:
Please enter a whole number between 1 and 10.

Step 5: Set Error Alert

1. Click the Error Alert tab.
2. Choose:
 - **Style:** Stop
 - **Title:** Invalid Input
 - **Error Message:**
Invalid input! The quantity must be a whole number between 1 and 10.

Step 6: Apply

- Click **OK** to apply the data validation.

This ensures accurate and controlled data entry for analysis.

Q6. Understand and apply fundamental text functions like LEFT, RIGHT, MID, and LEN. Extract the first 5 characters from the string "ExcelTipsAreGreat" using the LEFT function. Extract the last 4 characters from "DataAnalysis.xlsx" using the RIGHT function. Extract the substring "Tips" from "ExcelTipsAreGreat" using the MID function. Count the total number of characters in the string "Hello World!" using the LEN function. Create a formula to extract the middle 6 characters from "12345-67890-ABCDE".

Ans- Excel text functions help in extracting, counting, and cleaning text data.

- **LEFT Function**

The LEFT function extracts characters from the beginning of a text.

=LEFT("ExcelTipsAreGreat",5)

Result: Excel

- **RIGHT Function**

The RIGHT function extracts characters from the end of a text.

=RIGHT("DataAnalysis.xlsx",4)

Result: xlsx

- **MID Function**

The MID function extracts text from the middle of a string.

=MID("ExcelTipsAreGreat",6,4)

Result: Tips

- **LEN Function**

The LEN function counts the total number of characters in a text.

=LEN ("Hello World!")

Result: 12

- **Extract Middle Characters**

To extract the middle 6 characters from a string:

=MID("12345-67890-ABCDE",7,6)

Result: 67890-

- Q7. Understand how to combine text using CONCAT, TEXTJOIN, and the & operator.
- a.) Use CONCAT to combine "Hello" and "World" with a space in between.
 - b.) Combine "Apple", "Banana", and "Cherry" into a single string separated by commas using TEXTJOIN.
 - c.) Use the & operator to create the string "2025: Excel Functions" by combining "2025", ":", and "Excel Functions".
 - d.) Create a comma-separated list from the range A1:A5 using TEXTJOIN
 - e.) Combine first names in column A with last names in column B to create full names in column C

Ans- Excel provides different methods to combine text values such as CONCAT, TEXTJOIN, and the & operator.

a) Using CONCAT

To combine “Hello” and “World” with a space:

=CONCAT ("Hello", " ", "World")

Result: Hello World

b) Using TEXTJOIN

To combine “Apple”, “Banana”, and “Cherry” separated by commas:

=TEXTJOIN (", ", TRUE, "Apple", "Banana", "Cherry")

Result: Apple, Banana, Cherry

c) Using the & operator

To create the text “2025: Excel Functions”:

= "2025"&" : "&" Excel Functions"

Result: 2025: Excel Functions

d) Comma-separated list from range A1:A5

=TEXTJOIN (", ", TRUE, A1:A5)

This combines all values in cells A1 to A5, separated by commas.

e) Combine first name and last name

If first names are in column A and last names in column B:

=A1&" "&B1

This formula is entered in column C to create full names.

Q8. Understanding TODAY() and NOW()

- a.) What is the difference between TODAY() and NOW() in Excel? Provide an example of when you would use each function.
- b.) If cell A1 contains the date 2025-06-10, write a formula using TODAY() to determine how many days are left until that date
- c.) Write an Excel formula using NOW() to display the current date and time in the format MM/DD/YYYY HH:MM AM/PM.
- d.) If a cell contains =TODAY(), what will happen when the worksheet is reopened the next day? Explain
- e.) You want to store a static date (today's date) in a cell without it changing every day. What keyboard shortcut should you use

Ans- Understanding TODAY () and NOW () in Excel

a) Difference between TODAY () and NOW ()

TODAY () returns only the current date.

Example: Used to check how many days are left before a deadline.

NOW () returns the current date and time.

Example: Used to record the exact time when data is updated.

b) Days left until a future date

If cell A1 contains 2025-06-10, the formula is:

=A1-TODAY ()

This shows the number of days remaining until that date.

c) Display current date and time in required format

=TEXT (NOW (), "MM/DD/YYYY HH:MM AM/PM")

This displays the current date and time in the specified format.

d) What happens when the worksheet is reopened the next day?

If a cell contains =TODAY (), the date will automatically update to the new current date when the worksheet is opened again.

e) Store a static (non-changing) date

To enter today's date without it changing, use the keyboard shortcut

Ctrl +;